

# Frontend Documentation

## Project Overview

The frontend of the Automated Network Design and Deployment tool is a modern React application built for high performance, ease of use, and a streamlined network design workflow. It guides users through a sequential pipeline from initial requirements to final deployment configurations.

## Technical Stack & Tools

The application utilizes a cutting-edge frontend stack:

- **Framework:** React 19 - Utilizing the latest features for UI management and performance.
- **Build Tool:** Vite 8 - Providing an extremely fast development environment and optimized production builds.
- **Styling:** Tailwind CSS 4 - A utility-first CSS framework for rapid UI development and consistent design.
- **Routing:** React Router 7 - Manages the multi-step navigation flow.
- **State Management:** React Context API (**ProjectContext**) - Handles global project data across the design pipeline.
- **API Client:** Axios - For reliable communication with the backend services.
- **Content Rendering:** Marked - To render complex network documentation and summaries from Markdown.

## Design & Architecture

The UI follows a clean, professional “Enterprise Dashboard” aesthetic.

### Layout Structure

- **AppLayout:** The primary wrapper providing a persistent sidebar and top navigation bar across all views.
- **Sidebar:** Contextual navigation tracking the progress through the design steps.
- **TopBar:** Global actions and project metadata display.

### Design Principles

- **Modern UI:** Minimalist design with a focus on data clarity.
- **Responsiveness:** Fluid layouts that adapt to various screen sizes.
- **Component-Based:** Highly modular code using reusable UI components.

## Application Flow

The application implements a strict 7-step pipeline to ensure valid network designs:

1. **Dashboard:** Project overview and recent activity.
2. **Solution Type:** Selection of network architecture (e.g., Campus, Branch, Data Center).
3. **Requirements:** Input of user-specific constraints and scale needs.
4. **Proposed Design:** Visualization of the generated high-level architecture.
5. **Bill of Materials (BOM):** Automated list of required hardware and software licenses.
6. **Detailed Topology:** In-depth technical view of device connections.
7. **Deployment:** Generation of configuration files and deployment status.

## How to Run & Build

### Prerequisites

- Node.js (Latest LTS recommended)
- npm or yarn

### Installation

Navigate to the frontend source directory and install dependencies:

```
cd frontend/code  
npm install
```

### Development Server

Start the development server with Hot Module Replacement (HMR):

```
npm run dev
```

The application will typically be available at `http://localhost:5173`.

### Production Build

Create an optimized production bundle in the `dist` directory:

```
npm run build
```

### Preview Production Build

Locally preview the production build:

```
npm run preview
```

### Linting

Maintain code quality and style standards:

```
npm run lint
```